SMA PHYSICS — TARGET JEE MAIN

DPP-43: Rotational Motion: Centre of Mass of a Two-Particle System

Q01. Two particles of masses **2 kg** and **3 kg** are located along the x-axis at **x = 1 m** and **x = 6 m** respectively. The coordinate of their centre of mass is:

(A) 3.0 m
(B) 4.0 m
(C) 3.5 m
(D) 4.5 m

Q02. The distance between the carbon and oxygen atom in a **CO** molecule is **1.12 Å**. Taking the mass of carbon as **12 u** and oxygen as **16 u**, the distance of the centre of mass from the carbon atom is:

(A) 0.64 Å
(B) 0.48 Å
(C) 0.56 Å
(D) 0.72 Å

Q03. Two particles of masses **m_1 = 1 kg** and **m_2 = 3 kg** are initially separated by a distance **L**. If the mass **m_1** is shifted towards **m_2** by a distance of **6 cm**, by what distance must **m_2** be moved so that the position of the centre of mass remains unchanged?

(A) 2 cm towards **m_1**
(B) 2 cm away from **m_1**
(C) 18 cm towards **m_1**
(D) 6 cm away from **m_1**

Q04. Two bodies of masses **2 kg** and **4 kg** are moving with velocities **(2\hat{i - 7\hat{j + 3\hat{k) m/s** and **(-2\hat{i + 2\hat{j - \hat{k) m/s** respectively. The velocity of their centre of mass is:

(A) {1{3(-2\hat{i - 3\hat{j + \hat{k) m/s
(B) {1{3(-2\hat{i - 5\hat{j + 2\hat{k) m/s
(C) {1{6(-4\hat{i - 6\hat{j + 2\hat{k) m/s
(D) {1{3(-\hat{i - 3\hat{j + \hat{k) m/s

Q05. Two particles of masses **m** and **2m** are placed at positions **(0, 0)** and **(a, b)** respectively. If another particle of mass **m** is added at **(2a, 0)**, what is the new position of the centre of mass?

(A) ({a{2, {b{2)
(B) (a, {b{2)
(C) ({4a{3, {b{3)
(D) (a, {b{4)

Q06. A man of mass **50 kg** stands at one end of a stationary boat of mass **200 kg** and length **5 m** in still water. If the man walks to the other end of the boat, the displacement of the boat relative to the water is:

(A) 1.0 m
(B) 1.25 m
(C) 0.8 m
(D) 1.5 m

Q07. Two particles of masses **m_1** and **m_2** are placed on a smooth horizontal surface connected by a light spring. If a constant force **F** pulls **m_1** towards the right, the acceleration of the centre of mass of the system is:

(A) {F{m_1
(B) {F{m_1 + m_2
(C) {F{m_2
(D) Dependent on spring extension

Q08. Two masses **m_1 = 2 kg** and **m_2 = 8 kg** are released from rest under mutual gravitational attraction from an initial separation **d = 10 m**. When the separation between them reduces to **5 m**, the distance travelled by the **8 kg** mass is:

(A) 1.0 m
(B) 4.0 m
(C) 2.5 m
(D) 0.5 m

Q09. A projectile of mass **2m** is fired with initial velocity **u** at an angle θ to the horizontal. At the highest point of its trajectory, it explodes into two equal fragments. If one fragment falls vertically downwards with zero initial horizontal velocity, the horizontal distance of the other fragment from the launch point when it hits the ground is (**R = Range**):

(A) R
(B) 1.5 R
(C) 2 R
(D) 2.5 R

Q10. In a two-particle system with masses **m_1** and **m_2**, the distances of the particles from the centre of mass are **r_1** and **r_2** respectively. The ratio **r_1 / r_2** is always equal to:

(A) m_1 / m_2
(B) m_2 / m_1
(C) (m_2 / m_1)^2
(D) \sqrt{m_2 / m_1

Q11. Two particles of mass **3 kg** and **5 kg** have position vectors **(3\hat{i - 2\hat{j + \hat{k) m** and **(7\hat{i + 6\hat{j - 7\hat{k) m** respectively. The position vector of their centre of mass is:

(A) (5.5\hat{i + 3\hat{j - 4\hat{k) m
(B) (5.5\hat{i + 3\hat{j + 4\hat{k) m
(C) (5.5\hat{i - 3\hat{j - 4\hat{k) m
(D) (4\hat{i + 2\hat{j - 3\hat{k) m

Q12. Two blocks of masses **4 kg** and **6 kg** are connected by a compressed spring on a smooth table. When released from rest, the **4 kg** block moves to the left with an initial velocity of **9 m/s**. The velocity of the centre of mass of the system is:

(A) 0 m/s
(B) 3.6 m/s
(C) 6 m/s
(D) 5.4 m/s

Q13. A wooden plank of mass **M = 80 kg** and length **L = 6 m** rests on frictionless ice. A boy of mass **m = 40 kg** walking from one end stops at the center of the plank. The displacement of the boy relative to the ice is:

(A) 2.0 m
(B) 1.0 m
(C) 3.0 m
(D) 1.5 m

Q14. Two particles of masses **m** and **2m** moving with velocities **v_1** and **v_2** collide in 1D. The total kinetic energy in the Centre of Mass reference frame is given by:

(A) {1{3 m (v_1 - v_2)^2
(B) {2{3 m (v_1 - v_2)^2
(C) {1{2 m (v_1 - v_2)^2
(D) m (v_1 - v_2)^2

Q15. Two particles of mass **m_1 = 2 kg** and **m_2 = 3 kg** are connected by a rigid massless rod of length **1 m**. The moment of inertia of the system about an axis perpendicular to the rod and passing through its centre of mass is:

(A) 1.2 kg·m^2
(B) 0.6 kg·m^2
(C) 2.4 kg·m^2
(D) 1.5 kg·m^2

Q16. If external forces acting on a two-particle system sum to zero, which of the following quantities of the centre of mass MUST remain constant?

(A) Position vector
(B) Linear momentum
(C) Total kinetic energy
(D) Angular velocity

Q17. Two particles of masses **m_1 = 4 kg** and **m_2 = 1 kg** have initial velocities **\vec{v_1 = 2\hat{i m/s** and **\vec{v_2 = -3\hat{i m/s**. The velocity of particle 1 relative to the centre of mass is:

(A) 1.0\hat{i m/s
(B) 0.6\hat{i m/s
(C) 1.5\hat{i m/s
(D) -1.0\hat{i m/s

Q18. Two bodies of masses **m** and **M** (**$M > m$**) are connected by a string of length **L**. If the system rotates with angular velocity ω about their mutual centre of mass on a smooth table, the tension in the string is:

(A) {mM{M+m ω^2 L
(B) (M+m) ω^2 L
(C) {M-m{M+m ω^2 L
(D) {mM{M-m ω^2 L

Q19. Two particles of masses **$m_1 = 2$ kg** and **$m_2 = 3$ kg** are moving towards each other under mutual attractive force. At an instant when their velocities are **6 m/s** and **4 m/s** respectively, the speed of the centre of mass of the system is:

(A) 0 m/s
(B) 2.4 m/s
(C) 4.8 m/s
(D) 1.2 m/s

Q20. Two objects of masses **m_1** and **m_2** having initial velocities **u_1** and **u_2** accelerate at rates **a_1** and **a_2** along the same straight line. The acceleration of the centre of mass is zero if:

(A) $m_1 a_1 + m_2 a_2 = 0$
(B) $m_1 a_2 + m_2 a_1 = 0$
(C) $a_1 = a_2$
(D) $m_1 u_1 + m_2 u_2 = 0$

ANSWER KEY (DPP-43)

Q01:B	Q02:A	Q03:B	Q04:A	Q05:B	Q06:A	Q07:B	Q08:A	Q09:B	Q10:B
Q11:A	Q12:A	Q13:A	Q14:A	Q15:A	Q16:B	Q17:A	Q18:A	Q19:A	Q20:A